



Assessment

(SDLC- Software Testing)

Submitted From

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RISEUP LABS

- **Write 10 Test cases**

  https://docs.google.com/spreadsheets/d/11CKi1RZ8pViUbZ5bI1m_y260StIYZSUNNuhzhBP0G1I/edit?usp=sharing

- **Write a difference between Sanity testing and Regression testing**

Sanity Testing	Regression Testing
Sanity Testing is performed to check the stability of new functionality or code changes in the existing build.	Regression testing is performed to check the stability of all areas impacted by any functionality change or code change.
Sanity Testing is part of Regression Testing	Regression Testing is independent testing.
Sanity Testing is considered as a Surface Level Testing.	Regression Testing is not considered as a Surface Level Testing.
It examines few functionalities of the software.	It examines extended mostly all functionality of the software.
Sanity Testing does not use script.	Regression Testing uses Scripts.
Sanity Testing is often carried out manually.	Regression Testing is often preferred to continue with automation.
Performing Sanity Testing increases the product cost/budget cost.	Performing Regression Testing increases the product cost/budget cost.

- ## Annotate Exploratory testing, Happy path Testing, Unit Testing and Functional Testing

Exploratory testing: It is a testing approach where testers perform testing based on their knowledge, experience, and intuition also experience, rather than following predefined steps.

Happy path Testing: Happy-path testing is a type of software testing that uses known input and produces an expected output. Also referred to as golden-path or sunny-day testing, the happy-path approach is tightly scripted. It verifies that the system functions correctly under normal conditions without errors or exceptions.

Unit Testing: Unit testing is the process where you test the smallest functional unit of code. Software testing helps ensure code quality, and it's an integral part of software development. It ensures each unit behaves as expected and meets its design specifications.

Functional Testing: Functional testing is a type of software testing where the basic functionalities of an application are tested against a predetermined set of specifications. It involves checking both the functional and non-functional aspects of the application to ensure it performs as intended in various scenarios.

- ## Run a Login and Registration Automation of QA Brain with Selenium

Automation testing is a software testing technique that uses specialized tools and scripts to automatically execute test cases and verify the outcomes against expected results. This approach aims to increase the efficiency, effectiveness, and coverage of testing processes, while reducing the time and effort required for manual testing.

To automate login and registration functionality using Selenium for a website like QA Brain, we can use the Node.js runtime along with the selenium-webdriver package.

Here is the step-by-step procedure:

• **Install Node.js**: First of all we need to download and install Node.js from nodejs.org.

• **Create a New Node.js Project**: Then we need to open a terminal or command prompt, and need to create a new directory for our project, and initialize a new Node.js project:

```
mkdir qa-brain-automation
cd qa-brain-automation
npm init -y
```

Install Selenium WebDriver: Then we need to install the selenium-webdriver package:

```
npm install selenium-webdriver
```

• **Download the WebDriver**: Then we need to Download the WebDriver from the browser.

• **Create the Automation Script**: Then create a file named index.js:

```
const { Builder, By, Key, until } = require('selenium-webdriver');

async function register() {
    let driver = await new Builder().forBrowser('chrome').build();
    try {
        await driver.get('https://qabrain.com/register'); // Use the
correct URL
        await driver.findElement(By.name('username')).sendKeys('user1');
        await
driver.findElement(By.name('email')).sendKeys('user1@example.com');
        await driver.findElement(By.name('password')).sendKeys('pass567');
        await
driver.findElement(By.name('confirm_password')).sendKeys('pass567',
Key.RETURN);

        // Wait for the registration process to complete
        await driver.wait(until.titleIs('Registration Successful'),
10000);
    } finally {
        await driver.quit();
    }
}

async function login() {
    let driver = await new Builder().forBrowser('chrome').build();
    try {
        await driver.get('https://qabrain.com/login'); // Use the correct
URL
        await driver.findElement(By.name('username')).sendKeys('user1');
        await driver.findElement(By.name('password')).sendKeys('pass567',
Key.RETURN);

        // Wait for the login process to complete
        await driver.wait(until.titleIs('Dashboard'), 10000);
```

```
    } finally {
        await driver.quit();
    }
}

async function runTests() {
    await register();
    await login();
}

runTests();
```

• **Run the Script**: Then we need to run the script.

This script provides a basic example of automation testing.